```
# Check entailments
entails_R = all(R for P, Q, R in models)

entails_R_implies_P = all(
    (not R) or P for P, Q, R in models
)

entails_Q_implies_R = all(
```

Problems Output Debug Console Terminal Ports

```
PS C:\Users\CHANDANA\OneDrive\Desktop\New folder> python propositional.py
P Q R | Q->P | P->¬Q | Q|R | KB
-----------------------------------
False False False | True | True | False | False
False False True | True | True | True | True
False True False | False | True | True | False
False True True | False | True | True | False
True False False | True | True | False | False
True False True | True | True | True | True
True True False | True | False | True | False
True True True | True | False | True | False

Models of KB: [(False, False, True), (True, False, True)]

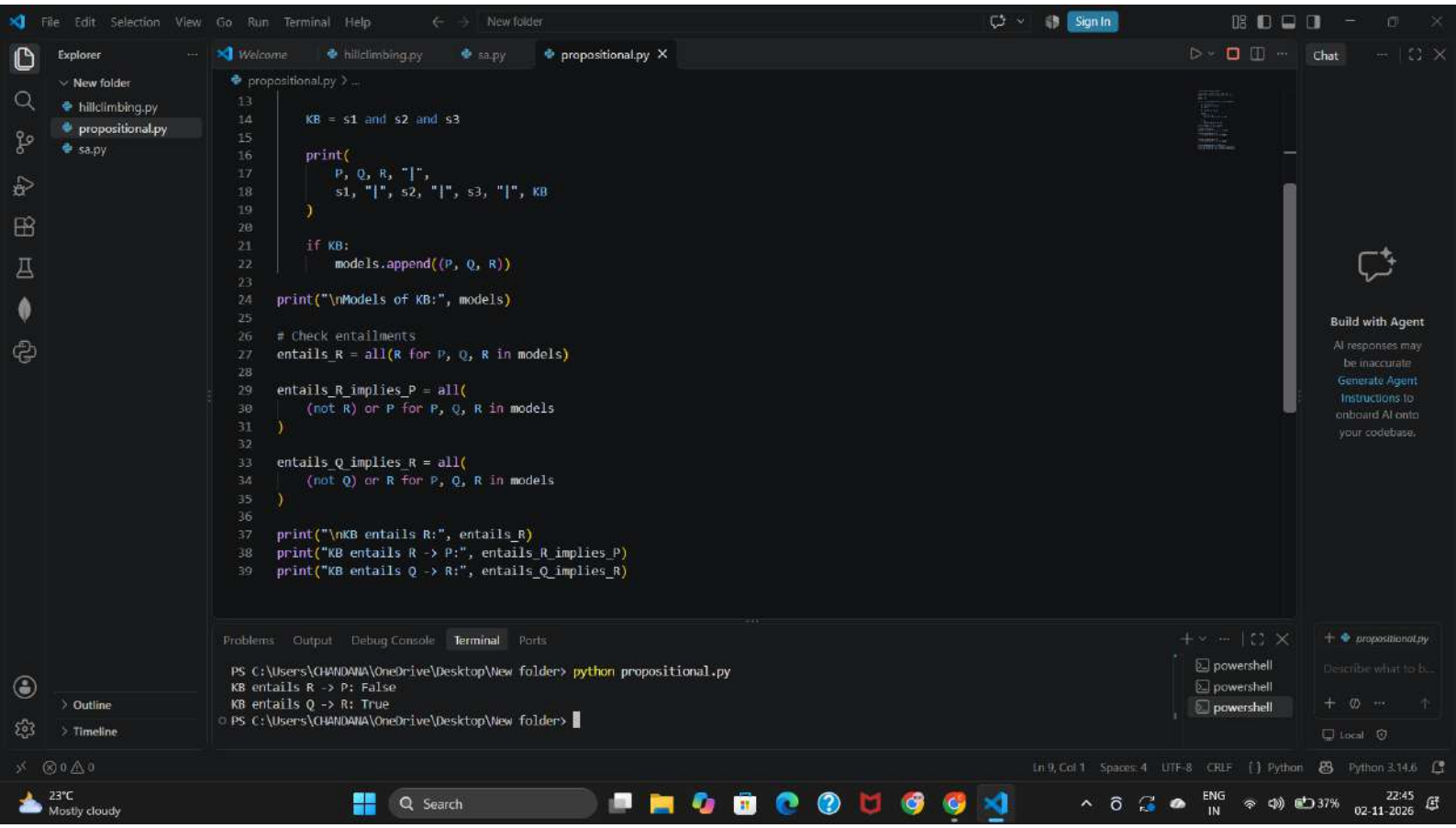
KB entails R: True
KB entails R -> P: False
KB entails Q -> R: True
PS C:\Users\CHANDANA\OneDrive\Desktop\New folder>
```

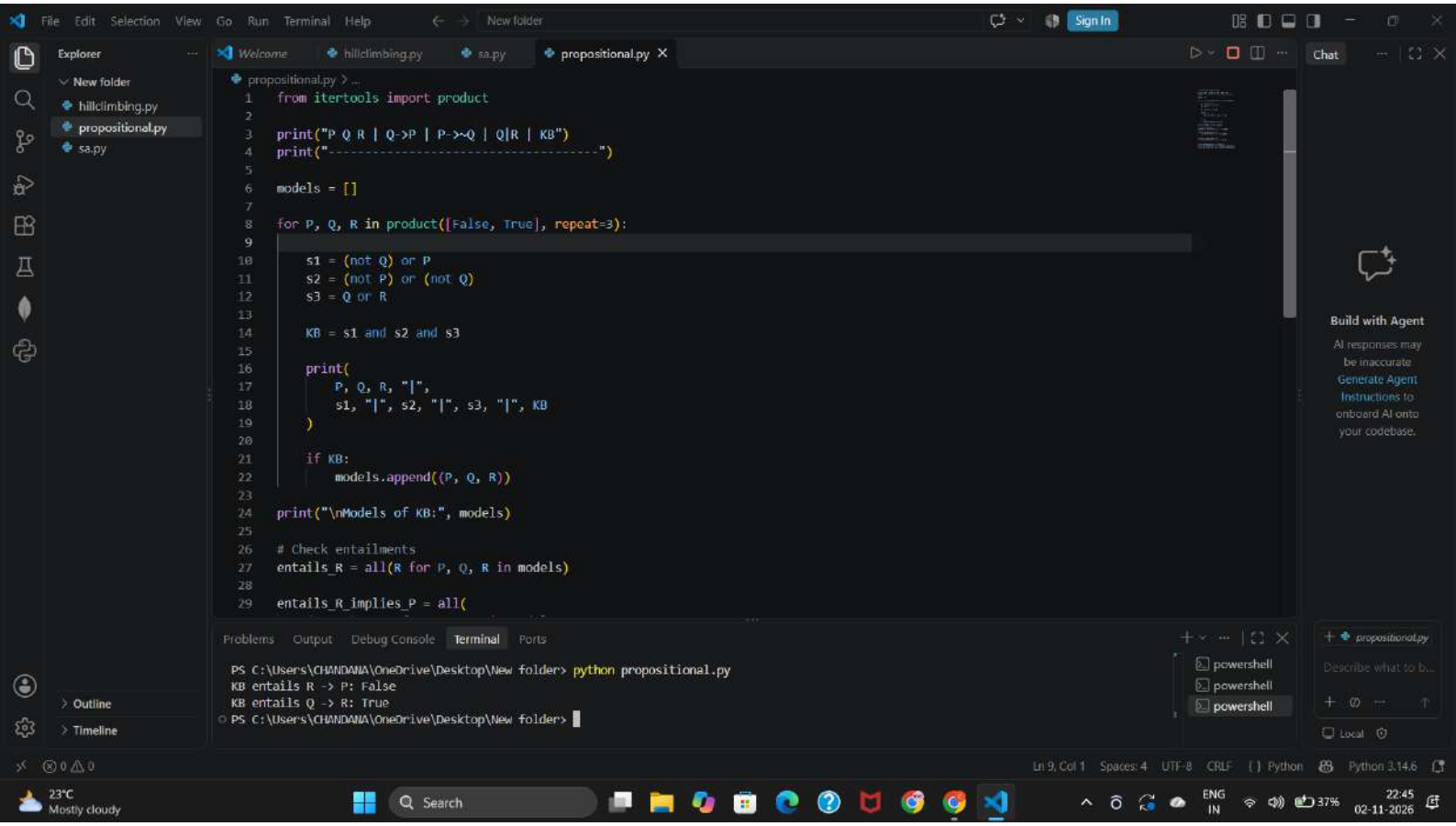
powershe
powershe
powershe

Ln 9, Col 1 Spaces: 4 UTF-8 CRLF { }

Search

ENG
IN


File Edit Selection View Go Run Terminal Help
New folder
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propositional.py
sa.py
Welcome
hillclimbing.py
sa.py
propositional.py
propositional.py > ...
KB = s1 and s2 and s3
print(
P, Q, R, "|",
s1, "|", s2, "|", s3, "|", KB
)
if KB:
models.append((P, Q, R))
print("\nModels of KB:", models)
Check entailments
entails_R = all(R for P, Q, R in models)
entails_R_implies_P = all(
(not R) or P for P, Q, R in models
)
entails_Q_implies_R = all(
(not Q) or R for P, Q, R in models
)
print("\nKB entails R:", entails_R)
print("KB entails R -> P:", entails_R_implies_P)
print("KB entails Q -> R:", entails_Q_implies_R)
Chat
Build with Agent
AI responses may be inaccurate
Generate Agent Instructions to onboard AI onto your codebase.
Problems Output Debug Console Terminal Ports
PS C:\Users\CHANDANA\OneDrive\Desktop\New folder> python propositional.py
KB entails R -> P: False
KB entails Q -> R: True
PS C:\Users\CHANDANA\OneDrive\Desktop\New folder>
powershell
powershell
powershell
Outline
Timeline
Ln 9, Col 1 Spaces: 4 UTF-8 CRLF { } Python Python 3.14.6
23°C
Mostly cloudy
Search
ENG
IN
37%
22:45
02-11-2026


File Edit Selection View Go Run Terminal Help
New folder
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Chat
propositional.py > ...
from itertools import product
print("P Q R | Q->P | P->~Q | Q|R | KB")
print("-----------------------------------")
models = []
for P, Q, R in product([False, True], repeat=3):
s1 = (not Q) or P
s2 = (not P) or (not Q)
s3 = Q or R
KB = s1 and s2 and s3
print(
P, Q, R, "|",
s1, "|", s2, "|", s3, "|", KB
)
if KB:
models.append((P, Q, R))
print("\nModels of KB:", models)
Check entailments
entails_R = all(R for P, Q, R in models)
entails_R_implies_P = all(
Build with Agent
AI responses may be inaccurate
Generate Agent Instructions to onboard AI onto your codebase.
Problems
Output
Debug Console
Terminal
Ports
PS C:\Users\CHANDANA\OneDrive\Desktop\New folder> python propositional.py
KB entails R -> P: False
KB entails Q -> R: True
PS C:\Users\CHANDANA\OneDrive\Desktop\New folder>
powershell
powershell
powershell
Outline
Timeline
Ln 9, Col 1
Spaces: 4
UTF-8
CRLF
Python
Python 3.14.6
23°C
Mostly cloudy
Search
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22:45
02-11-2026

```
64              if random.random() < probability:
65                  state = neighbor
66                  current_cost = neighbor_cost
```

Problems Output Debug Console **Terminal** Ports

```
PS C:\Users\CHANDANA\OneDrive\Desktop\New folder> python sa.py
Enter 8 queen positions (0-7): 0 4 7 5 2 6 1 3

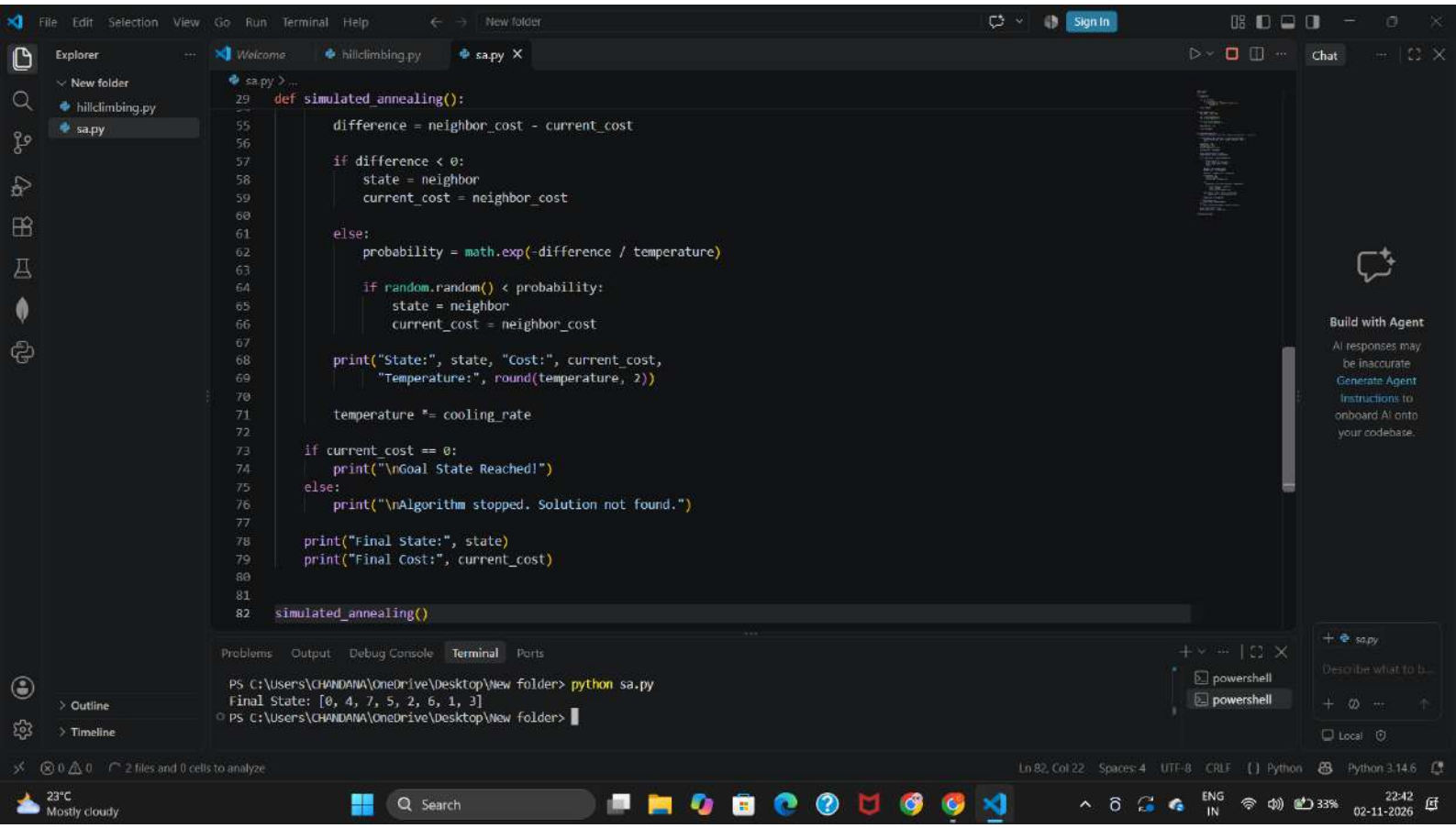
Initial State: [0, 4, 7, 5, 2, 6, 1, 3]
Initial Cost: 0

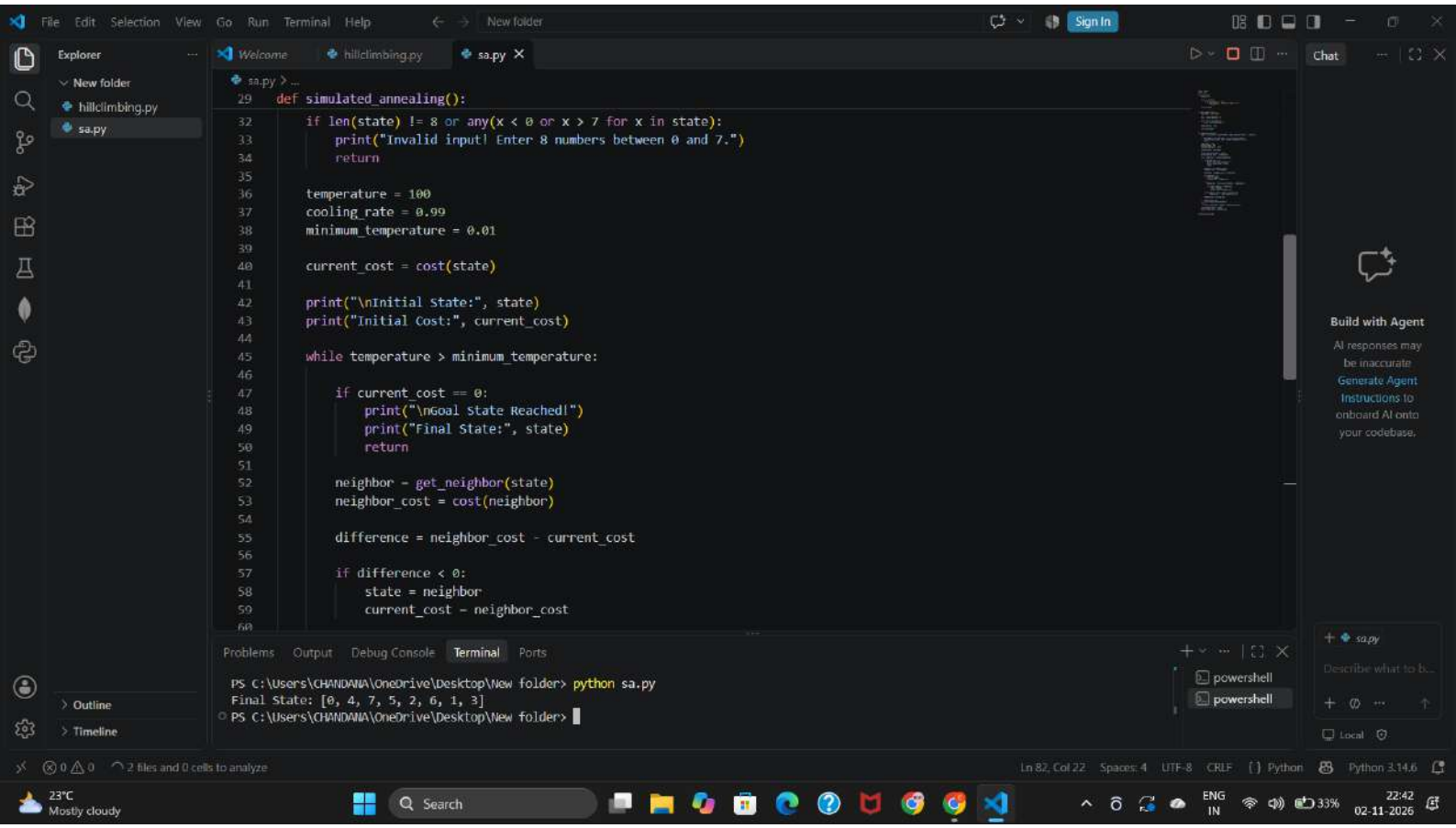
Goal State Reached!
Final State: [0, 4, 7, 5, 2, 6, 1, 3]
PS C:\Users\CHANDANA\OneDrive\Desktop\New folder>
```

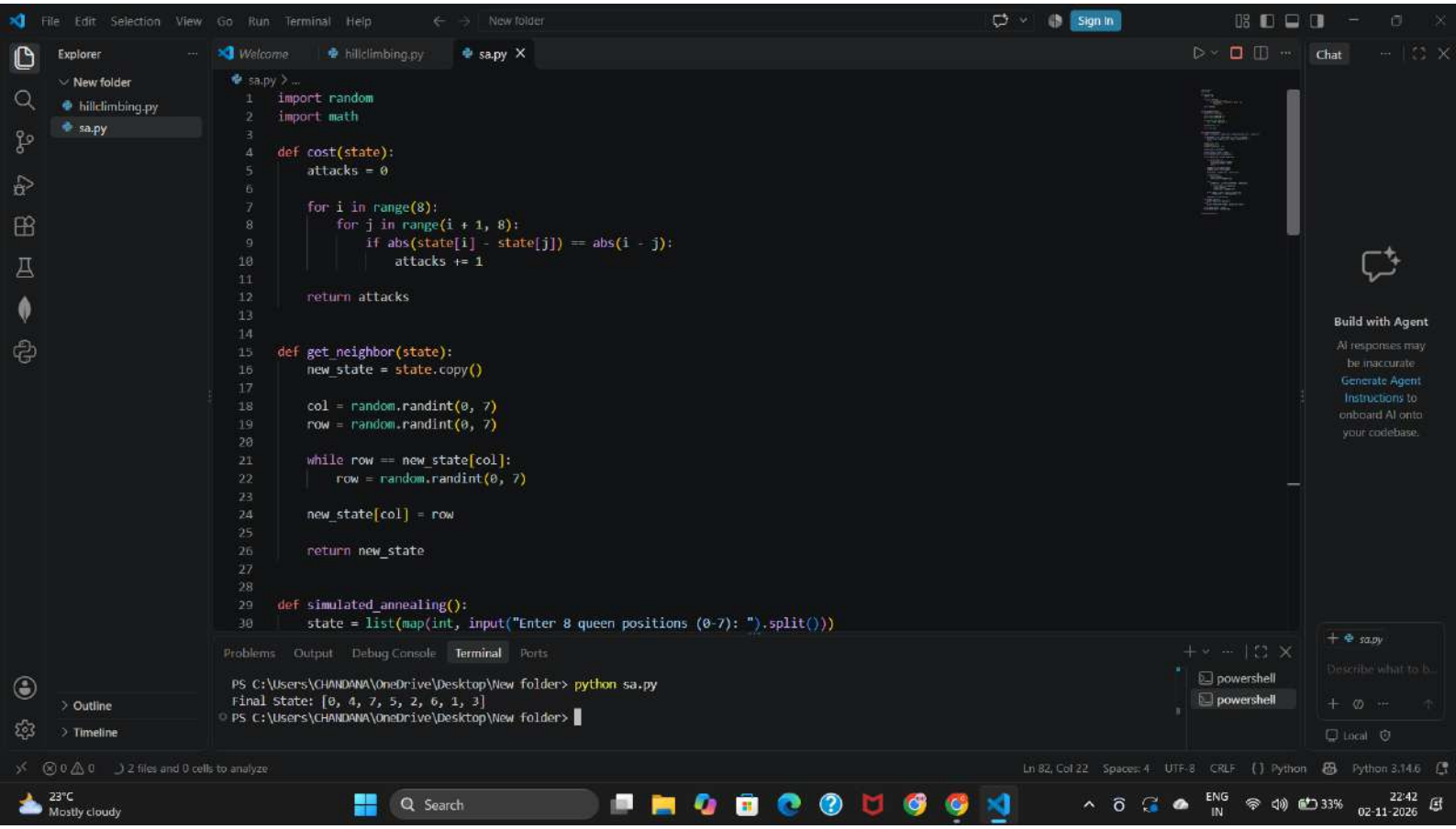
les and 0 cells to analyze

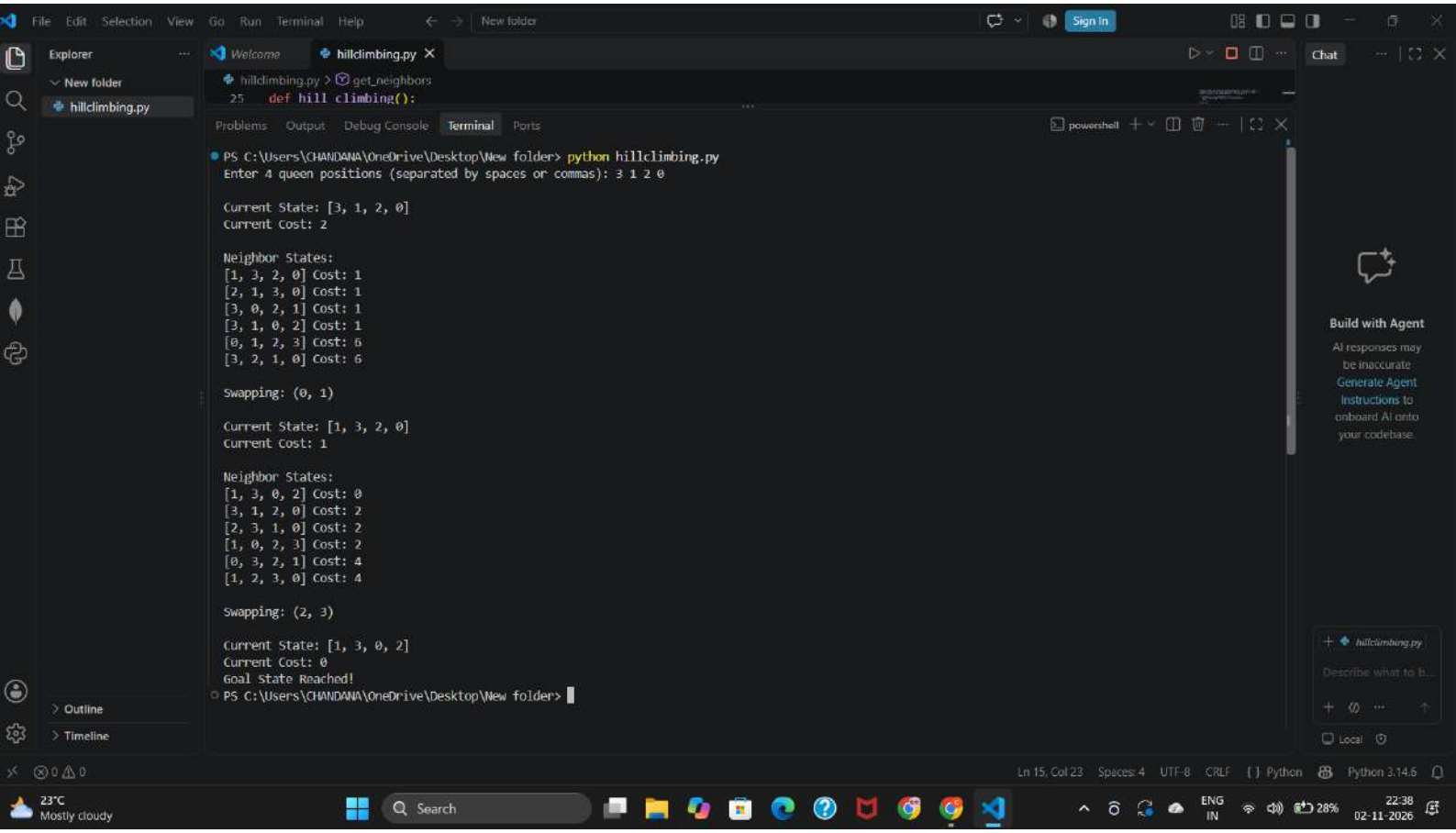
Ln 82, Col 22 Spaces: 4

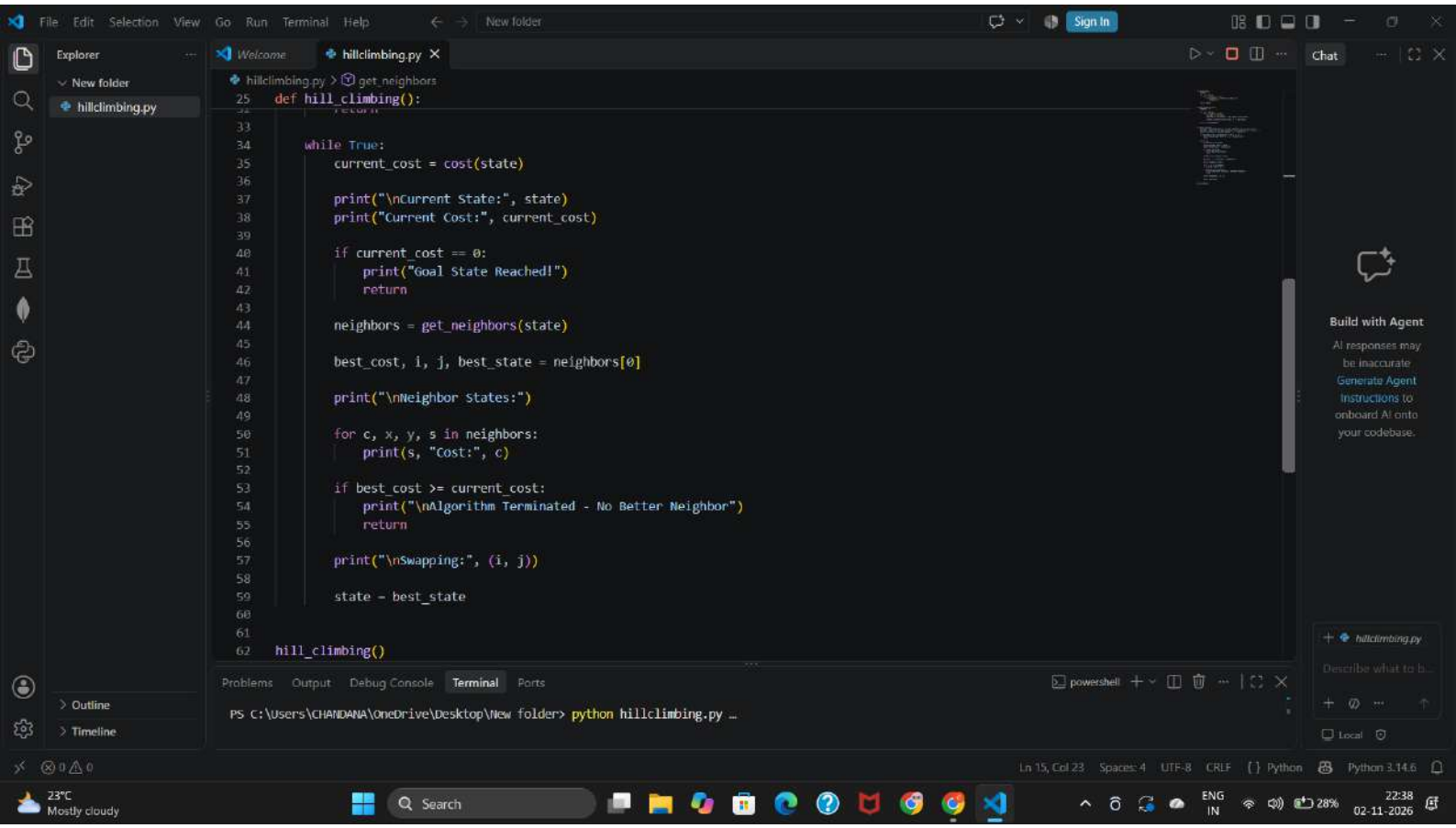


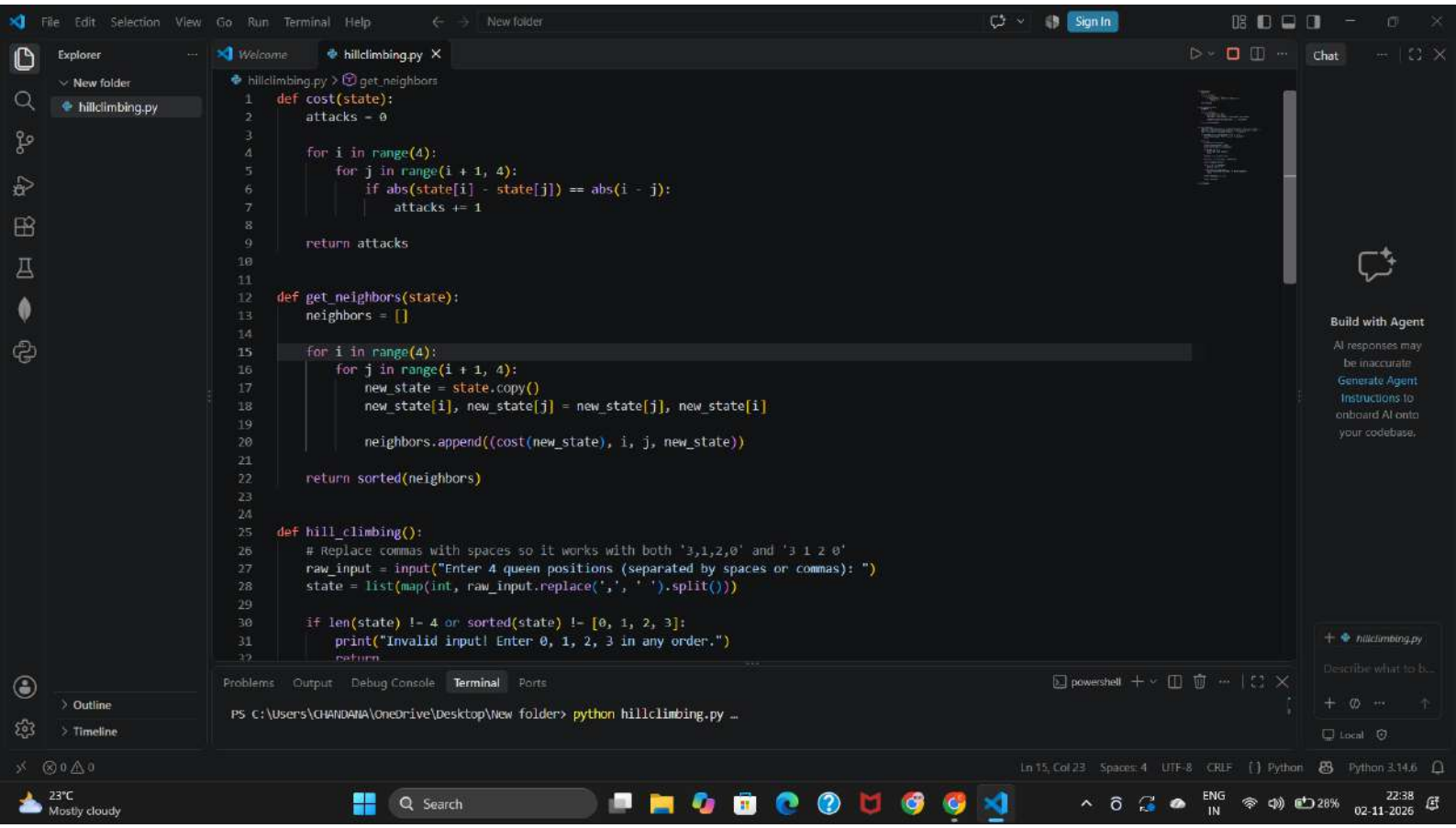

File Edit Selection View Go Run Terminal Help
New folder
Sign In
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hillclimbing.py
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Chat
sa.py > ...
29 def simulated_annealing():
55 difference = neighbor_cost - current_cost
56
57 if difference < 0:
58 state = neighbor
59 current_cost = neighbor_cost
60
61 else:
62 probability = math.exp(-difference / temperature)
63
64 if random.random() < probability:
65 state = neighbor
66 current_cost = neighbor_cost
67
68 print("State:", state, "Cost:", current_cost,
69 "Temperature:", round(temperature, 2))
70
71 temperature *= cooling_rate
72
73 if current_cost == 0:
74 print("\nGoal State Reached!")
75 else:
76 print("\nAlgorithm stopped. Solution not found.")
77
78 print("Final State:", state)
79 print("Final Cost:", current_cost)
80
81
82 simulated_annealing()
Build with Agent
AI responses may be inaccurate
Generate Agent Instructions to onboard AI onto your codebase.
Problems Output Debug Console Terminal Ports
PS C:\Users\CHANDANA\OneDrive\Desktop\New folder> python sa.py
Final State: [0, 4, 7, 5, 2, 6, 1, 3]
PS C:\Users\CHANDANA\OneDrive\Desktop\New folder>
powershell
powershell
Outline
Timeline
2 files and 0 cells to analyze
Ln 82, Col 22 Spaces: 4 UTF-8 CRLF Python Python 3.14.6
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Mostly cloudy
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22:42
02-11-2026


File Edit Selection View Go Run Terminal Help
New folder
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New folder
hillclimbing.py
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Chat
sa.py > ...
29 def simulated_annealing():
32 if len(state) != 8 or any(x < 0 or x > 7 for x in state):
33 print("Invalid input! Enter 8 numbers between 0 and 7.")
34 return
36 temperature = 100
37 cooling_rate = 0.99
38 minimum_temperature = 0.01
40 current_cost = cost(state)
42 print("\nInitial State:", state)
43 print("Initial Cost:", current_cost)
45 while temperature > minimum_temperature:
47 if current_cost == 0:
48 print("\nGoal State Reached!")
49 print("Final State:", state)
50 return
52 neighbor = get_neighbor(state)
53 neighbor_cost = cost(neighbor)
55 difference = neighbor_cost - current_cost
57 if difference < 0:
58 state = neighbor
59 current_cost = neighbor_cost
Build with Agent
AI responses may be inaccurate
Generate Agent Instructions to onboard AI onto your codebase.
Problems Output Debug Console Terminal Ports
PS C:\Users\CHANDANA\OneDrive\Desktop\New folder> python sa.py
Final State: [0, 4, 7, 5, 2, 6, 1, 3]
PS C:\Users\CHANDANA\OneDrive\Desktop\New folder>
powershell
powershell
Outline
Timeline
2 files and 0 cells to analyze
Ln 82, Col 22 Spaces: 4 UTF-8 CRLF Python Python 3.14.6
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File Edit Selection View Go Run Terminal Help
New folder
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New folder
hillclimbing.py
sa.py
Welcome
hillclimbing.py
sa.py
Chat
sa.py > ...
import random
import math
def cost(state):
attacks = 0
for i in range(8):
for j in range(i + 1, 8):
if abs(state[i] - state[j]) == abs(i - j):
attacks += 1
return attacks
def get_neighbor(state):
new_state = state.copy()
col = random.randint(0, 7)
row = random.randint(0, 7)
while row == new_state[col]:
row = random.randint(0, 7)
new_state[col] = row
return new_state
def simulated_annealing():
state = list(map(int, input("Enter 8 queen positions (0-7): ").split()))
Build with Agent
AI responses may be inaccurate
Generate Agent Instructions to onboard AI onto your codebase.
Problems
Output
Debug Console
Terminal
Ports
PS C:\Users\CHANDANA\OneDrive\Desktop\New folder> python sa.py
Final State: [0, 4, 7, 5, 2, 6, 1, 3]
PS C:\Users\CHANDANA\OneDrive\Desktop\New folder>
powershell
powershell
Outline
Timeline
Ln 82, Col 22
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File Edit Selection View Go Run Terminal Help
New folder
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New folder
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Welcome
hillclimbing.py
hillclimbing.py > get_neighbors
25 def hill climbing():
Problems Output Debug Console Terminal Ports
powershell
PS C:\Users\CHANDANA\OneDrive\Desktop\New folder> python hillclimbing.py
Enter 4 queen positions (separated by spaces or commas): 3 1 2 0
Current State: [3, 1, 2, 0]
Current Cost: 2
Neighbor States:
[1, 3, 2, 0] Cost: 1
[2, 1, 3, 0] Cost: 1
[3, 0, 2, 1] Cost: 1
[3, 1, 0, 2] Cost: 1
[0, 1, 2, 3] Cost: 6
[3, 2, 1, 0] Cost: 6
Swapping: (0, 1)
Current State: [1, 3, 2, 0]
Current Cost: 1
Neighbor States:
[1, 3, 0, 2] Cost: 0
[3, 1, 2, 0] Cost: 2
[2, 3, 1, 0] Cost: 2
[1, 0, 2, 3] Cost: 2
[0, 3, 2, 1] Cost: 4
[1, 2, 3, 0] Cost: 4
Swapping: (2, 3)
Current State: [1, 3, 0, 2]
Current Cost: 0
Goal State Reached!
PS C:\Users\CHANDANA\OneDrive\Desktop\New folder>
Chat
Build with Agent
AI responses may be inaccurate
Generate Agent Instructions to onboard AI onto your codebase.
Outline
Timeline
Ln 15, Col 23 Spaces: 4 UTF-8 CRLF Python Python 3.14.6
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28%
22:38 02-11-2026


File Edit Selection View Go Run Terminal Help
New folder
Sign In
Explorer
New folder
hillclimbing.py
Welcome
hillclimbing.py
Chat
hillclimbing.py > get_neighbors
def hill_climbing():
while True:
current_cost = cost(state)
print("\nCurrent State:", state)
print("Current Cost:", current_cost)
if current_cost == 0:
print("Goal State Reached!")
return
neighbors = get_neighbors(state)
best_cost, i, j, best_state = neighbors[0]
print("\nNeighbor States:")
for c, x, y, s in neighbors:
print(s, "Cost:", c)
if best_cost >= current_cost:
print("\nAlgorithm Terminated - No Better Neighbor")
return
print("\nSwapping:", (i, j))
state = best_state
hill_climbing()
Build with Agent
AI responses may be inaccurate
Generate Agent Instructions to onboard AI onto your codebase.
Problems
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PS C:\Users\CHANDANA\OneDrive\Desktop\New folder> python hillclimbing.py ...
Outline
Timeline
Ln 15, Col 23
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File Edit Selection View Go Run Terminal Help
New folder
Sign In
Explorer
New folder
hillclimbing.py
Welcome
hillclimbing.py
hillclimbing.py > get_neighbors
def cost(state):
attacks = 0
for i in range(4):
for j in range(i + 1, 4):
if abs(state[i] - state[j]) == abs(i - j):
attacks += 1
return attacks
def get_neighbors(state):
neighbors = []
for i in range(4):
for j in range(i + 1, 4):
new_state = state.copy()
new_state[i], new_state[j] = new_state[j], new_state[i]
neighbors.append((cost(new_state), i, j, new_state))
return sorted(neighbors)
def hill_climbing():
Replace commas with spaces so it works with both '3,1,2,0' and '3 1 2 0'
raw_input = input("Enter 4 queen positions (separated by spaces or commas): ")
state = list(map(int, raw_input.replace(',', ' ').split()))
if len(state) != 4 or sorted(state) != [0, 1, 2, 3]:
print("Invalid input! Enter 0, 1, 2, 3 in any order.")
Chat
Build with Agent
AI responses may be inaccurate
Generate Agent Instructions to onboard AI onto your codebase.
Problems Output Debug Console Terminal Ports
powershell
PS C:\Users\CHANDANA\OneDrive\Desktop\New folder> python hillclimbing.py ...
Outline
Timeline
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